The PDF's stated objective is to work with different datasets and classifiers and draw inferences about their behaviour.

A practical restatement is:

| Compare several classification methods across different kinds of data, then investigate how reducing the feature dimension using

| PCA changes the results.

There are two central comparisons.

Comparison A: Classifier comparison

For a fixed dataset:

GMM vs LDA vs KNN vs SVM

For TIDIGITS, also include DTW.

You are asking:

- Which method performs best?
- Which assumptions seem appropriate?
- Which methods struggle with the representation?
- What computational trade-offs appear?

Comparison B: Representation comparison

For a fixed dataset and classifier:

Original features vs PCA features

You are asking:

- Is all the original dimensionality useful?
- Does removing dimensions remove noise?
- Does PCA discard information needed for classification?
- Can you achieve similar accuracy with fewer features?

The assignment is not simply “get the highest accuracy.” It is also “explain the differences.”

---

8. Learning outcomes

By completing the assignment, you should be able to:

Learning outcome	What it means in practice
Represent different data types	Understand fixed feature vectors versus variable-length sequences

├──────────────────────────────────────┼──┤

| Understand classifier assumptions | Recognize when linear, local, probabilistic, or sequence methods are appropriate |

├──────────────────────────────────────┼──┤

| Apply dimensionality reduction | Fit PCA correctly and interpret retained variance |

├──────────────────────────────────────┼──┤

| Design fair comparisons | Use consistent splits, representations, and evaluation procedures |

├──────────────────────────────────────┼──┤

| Avoid leakage | Keep test information out of training and model selection |

├──────────────────────────────────────┼──┤

| Interpret performance | Explain class confusions and accuracy changes |

├──────────────────────────────────────┼──┤

| Understand computation | Recognize why some methods become slow in high dimensions or on large datasets |

├──────────────────────────────────────┼──┤

| Produce reproducible experiments | Record preprocessing, random seeds, parameters, and results |

└──────────────────────────────────────┴──────────────────────────────────────┘

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9. Your inputs, dataset by dataset

The following separates what we have actually inspected from what still needs confirmation.

9.1 MNIST

Available input

```text
Datasets/MNIST/TEAM_3.zip
```

The archive contains train, dev, and test folders, with class folders for digits 1, 8, and 9.

What still needs inspection

- Actual image format
- Image dimensions
- Number of samples per split

- Whether all entries are valid images

Likely representation

If images are the usual 28 × 28:

28 × 28 → 784

Each image becomes one vector:

$x_i \in \mathbb{R}^{784}$

But the actual dimensions should be verified rather than assumed.

Target output

Predict:

$y_i \in \{1,8,9\}$

9.2 Image Features

Available input

```text
```

Datasets/ImageFeaturesDataset/Dataset2/

```

Contains:

- train_data.csv, train_label.csv
- val_data.csv, val_label.csv
- test_data.csv, test_label.csv

The inspected feature rows contain 36 values.

### Representation

$X \in \mathbb{R}^{n \times 36}$

Each row already describes one image.

### Important implication

You are not necessarily expected to process raw images here. The feature-extraction stage appears to have been done for you.

### Target output

Predict each sample's supplied class label.

The complete set of labels and their semantic meanings still need inspection.
```

9.3 Synthetic

Available input

```text
Datasets/SyntheticData/data.pkl
Datasets/SyntheticData/data_plot.png
```

What still needs inspection

- Structure of the stored object
- Feature dimension
- Number of classes
- Sample counts
- Whether predefined splits exist

Do not assume that the synthetic dataset is two-dimensional or Gaussian merely because it is synthetic.

Target output

Predict the provided class labels.

If it has two features, decision-boundary plots would be especially useful.

9.4 TIDIGITS

Available input

```text
  Datasets/TIDIGITS/3.npz
  Datasets/TIDIGITS/7.npz
  Datasets/TIDIGITS/9.npz
```

Each archive contains 450 arrays. Inspected arrays have shape:

$T_i \times 39$

with variable T_i.

Likely interpretation

- Each array is one recording’s feature sequence.
- Each row is one time frame.
- Each frame contains 39 features.
- The file name indicates the digit class.

This interpretation should be confirmed against any supplied documentation.

Target output

Predict:

$y_i \in \{3,7,9\}$

What is not established yet

- Exact meaning of the 39 features
- Speaker identities
- Recording/session identities
- Official split information

Without speaker metadata, we cannot honestly claim a speaker-independent evaluation.

---

10. Tasks and outputs for each classifier

There are two kinds of outputs:

1. Model output: what the method produces for an individual sample.
2. Experiment output: what you report after evaluating many samples.

10.1 GMM

Input

For non-speech data:

fixed-length feature vectors and labels

For a frame-based speech implementation:

feature sequences and recording labels

Task

Fit one GMM per class.

For a new sample or recording, evaluate its score under every class model.

Model output

- Class log-likelihood scores
- A predicted label

For example:

```text
 Score for class 3: ...
 Score for class 7: ...
 Score for class 9: ...
 Prediction: 7
```

```

### Experiment output

- Accuracy
- Confusion matrix
- Number of mixture components
- Covariance type
- Regularization settings
- Original-versus-PCA comparison

### Main assumptions

- A finite Gaussian mixture can approximate each class's feature distribution.
- Training samples represent the variation seen during testing.
- Frame-based speech scoring commonly assumes conditional frame independence.

---

## 10.2 LDA

### Input

Fixed-length vectors and labels.

For speech, first construct a fixed-length recording representation.
```

Task

Estimate:

- Class means
- Shared covariance
- Class priors

Then apply the linear discriminant scores.

Model output

- Class scores or posterior estimates
- Predicted label

Experiment output

- Accuracy and confusion matrix
- Solver/regularization choices
- Original-versus-PCA results

Main assumptions

- Approximately Gaussian class-conditional distributions
- Shared covariance across classes
- Useful class separation can be expressed through linear boundaries

These are modelling assumptions, not facts you must prove true.

10.3 KNN

Input

Fixed-length labelled training vectors.

Task

Choose:

- Number of neighbours k
- Distance measure
- Optional distance weighting

For each test vector, find neighbours and vote.

Model output

- Neighbour identities/distances, if requested
- Predicted label
- Optional vote-based probability estimates

Experiment output

- Accuracy and confusion matrix

- k, distance metric, weighting rule

- Prediction runtime, if measured

- Original-versus-PCA comparison

Main assumption

Nearby samples in the chosen representation tend to have the same class.

10.4 SVM

Input

Fixed-length labelled feature vectors.

Task

Learn a maximum-margin classification rule.

Choose:

- Linear or nonlinear kernel

- C

- Kernel-specific parameters, such as RBF γ

Model output

- Decision scores
- Predicted label

SVM decision scores are not automatically probabilities.

Experiment output

- Accuracy and confusion matrix
- Kernel and hyperparameters
- Original-versus-PCA comparison
- Runtime, if measured

Main assumptions

The classes can be usefully separated in the original or kernel-induced space, with regularization controlling complexity.

It does not require Gaussian class distributions.

10.5 DTW

Input

Ordered feature sequences:

$$X_i \in \mathbb{R}^{T \times d}$$

and labelled reference sequences.

Task

Calculate alignment costs between a test sequence and reference sequences.

Use a documented rule, such as nearest-reference classification.

Model output

- Alignment cost
- Optional alignment path
- Predicted class from the reference rule

Experiment output

- Accuracy and confusion matrix
- Local frame-distance definition
- Reference-set selection
- Path constraints, if any
- Cost normalization rule, if any
- Original-versus-PCA comparison

- Prediction time, if measured

Main assumptions

- Same-digit recordings contain comparable feature patterns in roughly the same order.
- Timing differences can be handled through monotonic alignment.
- Local feature distances are meaningful.

---

11. What “with PCA” actually requires

For every relevant dataset–classifier pair, you create two branches.

Branch A: Without PCA

```text
  Input
    → appropriate preprocessing
    → classifier
    → predictions
    → evaluation
```

Branch B: With PCA

```text
```

Input

→ same appropriate preprocessing

→ PCA fitted on training data

→ classifier

→ predictions

→ evaluation

```

“Without PCA” does not mean “without preprocessing.”

Both branches may still need:

- Image normalization
- Feature scaling
- Speech summarization
- Missing-value handling, if required

PCA should be the meaningful difference between the paired experiments.

### Choosing K

The PDF asks for projections onto the top K PCA dimensions, but does not specify exact values in the text we read.

Possible documented choices include:

- A fixed K
- The smallest K retaining a chosen variance fraction
```

- A small set of K values compared using validation data

Do not select K using test accuracy.

For your two-day deadline, one sensible documented selection rule is more manageable than a large sweep. Confirm any additional

instructor requirements.

---

12. Speech needs an explicit representation decision

This is the biggest design issue in your assignment.

Option A: Fixed-length summary

For a recording with 39 features per frame, calculate:

- 39 feature means
- 39 feature standard deviations

Concatenate them:

$s_i \in \mathbb{R}^{78}$

This can be used with LDA, ordinary KNN, and SVM.

What is retained?

Average feature values and their variability.

What is lost?

Temporal order.

Two recordings with reordered frames have the same mean/std summary.

Option B: Frame sequences

Keep:

$T \times 39$

i

$X_i \in \mathbb{R}$

Use these for:

- Frame-based GMM scoring
- DTW

For PCA:

$T_i \times 39 \rightarrow T_i \times K$

This retains sequence length and ordering.

Fairness caveat

If the methods use different speech representations, your comparison is between complete pipelines, not just isolated algorithms.

For example, better DTW performance might reflect the value of temporal information, not only the DTW rule itself.

State that clearly in the report.

---

13. Your complete experiment matrix

Dataset	Models	Versions	Configurations
MNIST	GMM, LDA, KNN, SVM	Original and PCA	8
Image Features	GMM, LDA, KNN, SVM	Original and PCA	8

| Synthetic | GMM, LDA, KNN, SVM | Original and PCA | 8 |

| TIDIGITS | GMM, LDA, KNN, SVM, DTW | Original and PCA | 10 |

| Total | | | 34 |

One GMM classification configuration may contain several class-specific GMMs internally.

Likewise, a multiclass SVM may contain multiple binary classifiers internally.

Those internal models do not change your 34-configuration count.

---

14. What should be reported?

14.1 Strictly explicit in the assignment

From the PDF:

1. Build the listed classifiers for the relevant datasets.

2. Compare classification accuracies.

3. Comment on the results.

4. Apply PCA and repeat using the top K dimensions.

14.2 Strongly recommended supporting outputs

These make your comparison convincing, even though the brief does not explicitly demand every item:

- Dataset summary
- Preprocessing description
- Split description
- Model hyperparameters
- PCA dimensions and retained variance
- Confusion matrices
- Training/prediction times
- Reproducible code and random seeds
- Limitations

Main results table

Create one row per configuration:

Dataset	Method	PCA?	Input dim.	Final dim.	Test accuracy	Notes
MNIST	KNN	No	d	d	measured	k = …

MNIST	KNN	Yes	d	K	measured	variance retained
…	…	…	…	…	…	…

Use actual measurements. Do not fill expected or invented accuracies.

---

15. How to evaluate correctly

Accuracy

$$\text{Accuracy} = \frac{\text{correct predictions}}{\text{total predictions}}$$

If 90 out of 100 samples are correct:

Accuracy = 90%

Confusion matrix

For classes 3,7,9, an illustrative matrix is:

$$M = \begin{pmatrix} 28 & 1 & 1 \\ 2 & 26 & 2 \\ 0 & 3 & 27 \end{pmatrix}$$

Using rows as true classes and columns as predicted classes:

- 28 true 3s were predicted as 3.
- One true 3 was predicted as 7.
- One true 3 was predicted as 9.

Accuracy is:

$$\frac{28+26+27}{90} = 90\%$$

These are example numbers, not your dataset's results.

The matrix reveals which errors are occurring, not merely how many.

---

16. What results should you expect?

You should expect patterns to investigate, not predetermined winners.

Possible observation	Possible interpretation—not a guaranteed explanation
LDA performs well	Linear separation and shared covariance may be reasonable approximations
GMM outperforms LDA	Multiple modes or more flexible class densities may matter
KNN improves with PCA	Removed dimensions may have been noisy or harmful to distances
Accuracy falls after PCA	Discarded directions may contain class information
SVM performs well on image vectors	The learned boundary may suit the representation

| DTW performs well on speech | Temporal alignment may help distinguish digit patterns |

| PCA reduces runtime with little accuracy loss | The original representation may contain substantial redundancy |

Do not automatically claim causation.

For example:

| “PCA improved KNN accuracy, consistent with the possibility that removing redundant or noisy directions improved neighbour

| relationships.”

That is more defensible than:

| “PCA proves that all removed dimensions were noise.”

---

17. Assumptions and decisions you must document

These are separate from each classifier’s mathematical assumptions.

Dataset assumptions

- Supplied labels are correct.
- File names identify speech classes, if confirmed.
- Training data is reasonably representative of test data.
- Existing splits should be respected unless instructed otherwise.

Experimental decisions

- How missing splits are created
- Whether class proportions are preserved
- Whether a training subset is used
- How features are scaled
- How speech is converted into fixed vectors
- How K is chosen
- Which hyperparameters are tried
- Which DTW references are used

Evaluation safeguards

- Fit scaling on training data only.
- Fit PCA on training data only.
- Train models on training data only.
- Select settings using validation data.
- Keep test data out of model selection.
- Use the same split for comparable experiments.

For speech, split recordings before pooling training frames for PCA or GMM fitting. Do not randomly split frames from the same

recording across training and test sets.

---

18. What does “finished” mean?

For each of the 34 configurations, you should be able to answer:

1. What was the input representation?
2. What preprocessing was applied?
3. What did the model learn or store?
4. Which hyperparameters were used?
5. Which samples were used for training, validation, and testing?
6. What labels did it predict?
7. What accuracy did it achieve?
8. How did PCA change its performance?
9. What limitations affect the interpretation?

Final mental picture

Your assignment is an investigation of this relationship:

[Performance = f(dataset, representation, classifier, hyperparameters, evaluation design)]

You are not simply testing five algorithm names.

You are testing how different modelling assumptions interact with different data representations—and whether PCA helps or hurts

those interactions.